

Photosynthesis – Mechanisms, Efficiency, and Relevance for Renewable Energy Systems

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Abstract—Photosynthesis is the biological process by which plants and other organisms convert sunlight, water, and carbon dioxide into chemical energy stored as biomass. Although photosynthesis and photovoltaic (PV) technology share the same energy source – sunlight – they differ fundamentally in their conversion mechanisms, efficiencies, and outputs. This paper examines the core mechanisms of photosynthesis, including the light-dependent reactions governed by Photosystem I and II and the light-independent Calvin cycle, with a focus on energy conversion from an engineering perspective. Efficiency limits and the key loss mechanisms that reduce solar-to-biomass conversion to roughly 1% in field conditions and up to 4% under optimal controlled conditions are analyzed and contrasted with commercial silicon PV modules achieving 18–20%. Finally, the emerging field of artificial photosynthesis is discussed as a complementary technology for producing solar fuels and renewable chemical feedstocks.

Index Terms—photosynthesis, photovoltaics, solar energy, energy conversion efficiency, artificial photosynthesis, renewable energy, Z-scheme, Calvin cycle

I. INTRODUCTION

The global transition toward renewable energy systems demands a broad understanding of natural and engineered solar energy conversion processes. Photosynthesis – the mechanism by which plants, algae, and cyanobacteria harness sunlight – represents the most widespread solar energy converter on Earth. It has sustained nearly all life on the planet for billions of years and currently underpins biomass energy, an important pillar of the renewable energy mix.

From the perspective of electrical engineering and energy systems, photosynthesis is of interest not only as a biological phenomenon but as an energy conversion system that can be analyzed, benchmarked, and potentially replicated. Both photosynthesis and photovoltaic (PV) technology convert electromagnetic radiation into a storable form of energy, yet their underlying principles, efficiencies, and outputs differ substantially.

This paper is structured as follows. Section II provides an overview of the two stages of photosynthesis. Section III examines the energy conversion mechanisms in detail, focusing on Photosystem I (PSI) and Photosystem II (PSII). Section IV analyzes the theoretical maximum efficiency and the main loss mechanisms. Section V compares photosynthesis to PV technology. Section VI discusses artificial photosynthesis as a pathway toward solar fuels. Section VII concludes the paper.

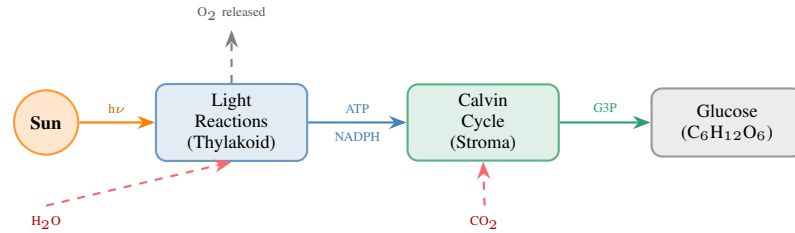


Fig. 1. Overview of the two-stage photosynthesis process. Light reactions in the thylakoid membrane convert sunlight, water, and ADP/NADP⁺ into ATP, NADPH, and O₂. The Calvin cycle uses these energy carriers to fix CO₂ into glucose.

II. FUNDAMENTALS OF PHOTOSYNTHESIS

Photosynthesis takes place in two sequential stages: the light-dependent reactions and the light-independent reactions (the Calvin cycle). Together, they convert solar photons, water, and carbon dioxide into glucose and oxygen according to the overall reaction:

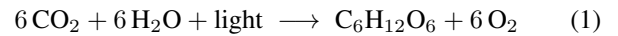


Fig. 1 shows the two-stage structure of photosynthesis and the flow of energy carriers between them.

A. Light-Dependent Reactions

The light-dependent reactions occur in the thylakoid membranes of chloroplasts. Their primary outputs are adenosine triphosphate (ATP) and nicotinamide adenine dinucleotide phosphate (NADPH), which serve as energy carriers for the subsequent Calvin cycle. Oxygen is released as a byproduct of water splitting [1].

B. Calvin Cycle

The Calvin cycle (also called the light-independent reactions) takes place in the stroma of the chloroplast. Using the ATP and NADPH generated in the light-dependent stage, carbon dioxide is fixed by the enzyme RuBisCO and reduced to form glyceraldehyde-3-phosphate (G3P), a precursor to glucose and other organic molecules. The Calvin cycle does not directly require light but depends entirely on the products of the light reactions [2].

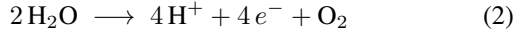
III. ENERGY CONVERSION MECHANISMS

A. The Z-Scheme

The electron flow through the light-dependent reactions is described by the *Z-scheme*, named for its characteristic shape when electron energy is plotted against the reaction sequence. Electrons are boosted twice by light absorption – once in PSII and once in PSI – allowing them to reach a sufficiently high energy level to reduce NADP^+ to NADPH [3]. Fig. 2 illustrates this energy diagram.

B. Photosystem II (PSII)

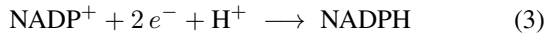
PSII contains a reaction-center chlorophyll pair known as P680, which absorbs photons at a wavelength of approximately 680 nm. Upon light absorption, one electron is excited to a higher energy state and transferred to an electron acceptor, initiating the electron transport chain. The electron deficit in P680 is replenished by the oxidative splitting of water molecules (photolysis), which releases molecular oxygen and protons:



As electrons travel through the chain of protein complexes between PSII and PSI (plastoquinone, the cytochrome b6f complex, and plastocyanin), their energy is used to pump protons across the thylakoid membrane, building a proton gradient. This gradient drives ATP synthase, producing ATP from ADP and inorganic phosphate – a mechanism directly analogous to the proton-motive force used in mitochondrial respiration [4].

C. Photosystem I (PSI)

By the time electrons reach PSI, they have lost a significant portion of their initial energy during transport. PSI's reaction center, P700, absorbs photons at approximately 700 nm and provides a second energy boost to re-excite these electrons. The re-energized electrons are ultimately transferred via ferredoxin to NADP^+ reductase, which catalyzes the reduction of NADP^+ to NADPH [1], [2]:



In addition to linear electron flow, PSI can engage in cyclic electron flow, recycling electrons back to the cytochrome b6f complex to produce additional ATP without generating NADPH . This regulatory mechanism allows the cell to balance the ATP/ NADPH ratio according to metabolic demand [5].

D. Energy Summary

The net result of the light-dependent reactions can be summarized as the conversion of photon energy into the chemical bond energy of ATP and NADPH . Four photons are required per electron pair extracted from water; since two electrons are needed to reduce one NADP^+ , and the Calvin cycle consumes three ATP and two NADPH per CO_2 fixed, the total photon cost per molecule of CO_2 fixed is approximately eight photons [6].

IV. EFFICIENCY ANALYSIS

A. Theoretical Maximum Efficiency

The theoretical maximum efficiency of oxygenic photosynthesis under ideal assumptions is often quoted at approximately 13% of total solar radiation [7]. When only photosynthetically active radiation (PAR, wavelengths 400–700 nm) is considered, the ceiling rises to roughly 9.4% for C3 plants and approximately 10.8% for C4 plants [6]. More conservative estimates for real plants under standard conditions yield practical upper limits of about 4.6% (C3) and 6% (C4) [7], [8].

B. Key Loss Mechanisms

Real-world photosynthetic efficiency falls far below these theoretical limits due to a cascade of loss mechanisms:

- **Spectral losses:** Plants can only use radiation in the PAR range. Infrared and ultraviolet radiation, which together constitute a substantial fraction of the solar spectrum, are not absorbed by chlorophyll and are therefore unavailable for photochemistry [6].
- **Reflection and transmission:** A fraction of incident PAR is reflected by the leaf surface or transmitted through the leaf without being absorbed [9].
- **Thermodynamic losses in charge separation:** When a photon excites an electron in P680 or P700, only a portion of the photon energy is stored in the excited state; the remainder is lost as heat due to rapid internal conversion and vibrational relaxation [9].
- **Electron transport chain losses:** As electrons flow downhill through the transport chain, a portion of their energy is dissipated as heat rather than being stored in the proton gradient. This represents an intrinsic thermodynamic cost of the system [10].
- **Photorespiration:** In C3 plants, the enzyme RuBisCO can mistakenly bind O_2 instead of CO_2 in a wasteful competing reaction. This process, called photorespiration, consumes ATP and NADPH without producing useful carbon compounds and can reduce net photosynthetic efficiency by 25% or more under warm, high-oxygen conditions [11].
- **Non-photochemical quenching (NPQ):** When light intensity exceeds the capacity of the Calvin cycle, excess absorbed energy is deliberately dissipated as heat or fluorescence to protect the photosynthetic machinery from oxidative damage. While essential for plant survival, NPQ reduces the fraction of absorbed light that contributes to photochemistry [9], [10].
- **Respiration and maintenance costs:** A portion of the fixed carbon is consumed by cellular respiration and other metabolic processes, reducing the net biomass accumulation available as an energy output [12].

The combined effect of these losses is that typical crop plants achieve a seasonal average solar-to-biomass efficiency of only about 1%, even under favorable field conditions [13].

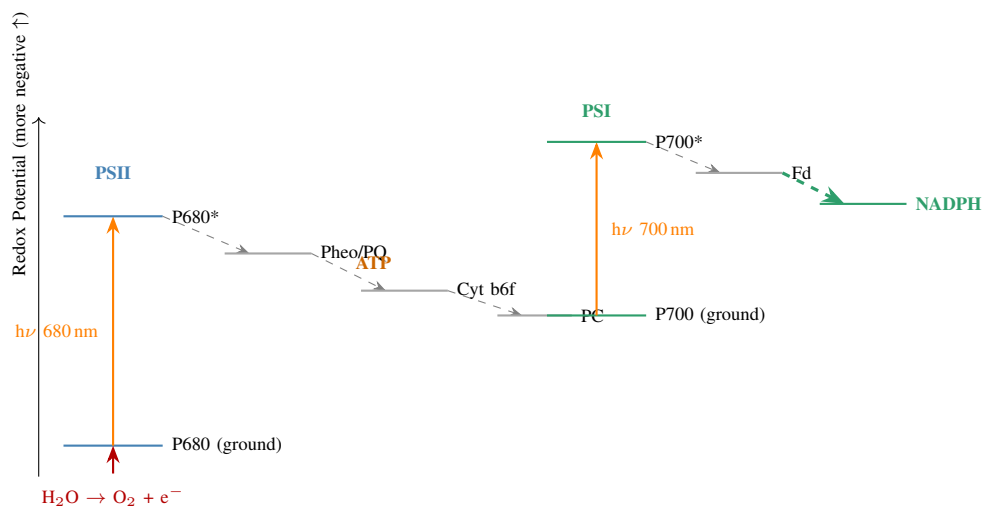


Fig. 2. Simplified Z-scheme energy diagram. Electrons are excited twice by photon absorption (orange arrows), flow through the electron transport chain generating ATP, and ultimately reduce NADP^+ to NADPH. Water is oxidized at PSII to replenish electrons.

TABLE I
SOLAR ENERGY CONVERSION EFFICIENCY COMPARISON

System	Efficiency	Notes
Crop photosynthesis	~1% (annual)	Field-scale, solar-to-biomass [13]
C3 plants (optimal)	~3.5%	Controlled conditions [15]
C4 plants (optimal)	~4.3%	Controlled conditions [15]
Commercial Si PV	18–20%	Module efficiency [15]
PV + electrolysis	10–12%	Solar-to-hydrogen [13]

V. COMPARISON WITH PHOTOVOLTAIC TECHNOLOGY

Table I summarizes the efficiency values for photosynthesis and competing solar energy conversion technologies.

PV cells convert photons directly into electric current via the photovoltaic effect, bypassing the many biological conversion steps that introduce losses in photosynthesis. A silicon PV cell absorbs a broader portion of the solar spectrum, operates without the overhead of cellular maintenance, and does not engage in competing reactions analogous to photorespiration [16]. Fig. 3 illustrates the efficiency gap visually.

However, a direct comparison requires acknowledging fundamental differences in outputs. PV produces electricity – a high-quality, immediately usable energy form – while photosynthesis produces biomass, a chemical energy carrier that can be stored, transported, and used as a feedstock for materials and fuels. PV-driven electrolysis (splitting water using PV-generated electricity) achieves approximately 10–12% annual solar-to-hydrogen efficiency under realistic conditions, still significantly higher than natural photosynthesis, but at considerably higher infrastructure cost [13], [14].

The two systems are therefore best understood as complementary rather than competing: PV is superior for direct electricity generation per unit area, while photosynthesis offers self-replication, self-repair, and the direct synthesis of complex

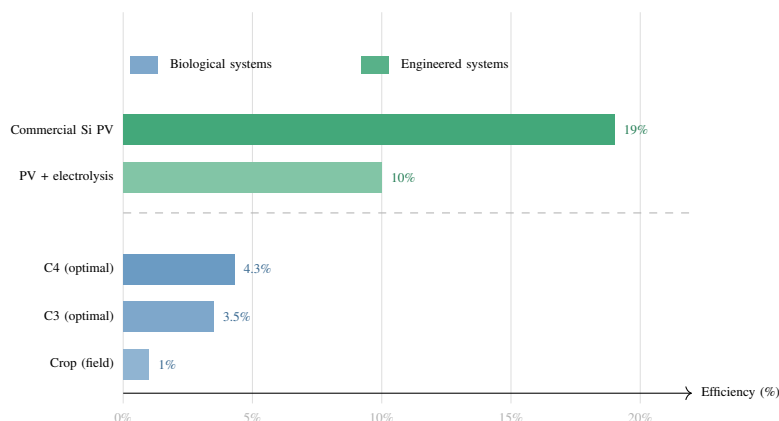


Fig. 3. Solar energy conversion efficiency comparison. Biological systems (bottom group) achieve 1–4%, while commercial silicon PV reaches ~19%. PV-driven electrolysis for H_2 production achieves ~10%.

organic molecules – capabilities that no current engineered system can replicate economically at scale [16].

VI. ARTIFICIAL PHOTOSYNTHESIS

A. Concept and Motivation

Artificial photosynthesis aims to replicate the core function of natural photosynthesis in engineered systems: using sunlight, water, and CO_2 to produce chemical fuels and feedstocks, without the biological overhead. The primary products of interest are hydrogen (via water splitting) and carbon-based fuels such as methanol or formic acid (via CO_2 reduction) [17], [18].

B. Current State of Research

Research in artificial photosynthesis spans several approaches, including photoelectrochemical (PEC) cells, molecular catalysts, photocatalytic suspensions, and semi-artificial

systems that incorporate isolated biological components such as enzymes or reaction center proteins into synthetic frameworks. Recent work has explored materials including perovskites, quantum dots, two-dimensional materials, and metal-organic frameworks as light absorbers and catalysts [19], [20].

Despite significant progress, the field remains predominantly at the laboratory and prototype stage. The key technical challenges are low solar-to-fuel efficiency, poor long-term stability of catalysts and photoabsorbers, slow reaction kinetics for multi-electron processes such as water oxidation, and high manufacturing costs for scalable systems [18], [21].

C. Relevance for Energy Systems

Artificial photosynthesis is strategically relevant for sectors that are difficult to electrify directly, including aviation, shipping, long-haul transport, and industrial chemistry (e.g., ammonia synthesis). In these applications, energy-dense liquid or gaseous fuels are more practical than batteries or direct electricity, and artificial photosynthesis offers a route to producing these fuels from sunlight in a carbon-neutral cycle [16], [22].

In the near term, the most realistic pathway is likely the integration of PV-generated electricity with electrochemical synthesis, effectively decoupling the light-capture and catalytic steps. Longer-term, fully integrated photoelectrochemical devices that mimic the spatial organization of the chloroplast – combining light absorption, charge separation, and catalysis in a single compact unit – represent the ultimate engineering goal [17], [18].

VII. CONCLUSION

Photosynthesis is a sophisticated, multi-stage solar energy conversion system that has been optimized by evolution for biological survival rather than maximum energy yield. Its core mechanisms – the Z-scheme electron transport through PSII and PSI, proton-gradient-driven ATP synthesis, and CO₂ fixation in the Calvin cycle – provide an elegant framework for understanding biological solar energy conversion from an engineering perspective.

When benchmarked against photovoltaic technology, natural photosynthesis is substantially less efficient: typical field efficiencies of 1% compare unfavorably with commercial silicon PV modules at 18–20%. The gap arises from spectral limitations, thermodynamic losses, photorespiration, and the metabolic overhead of maintaining a living system.

Nevertheless, photosynthesis offers unique capabilities – self-repair, self-replication, and direct production of complex organic molecules – that make it irreplaceable in the current energy landscape. Artificial photosynthesis seeks to bridge the gap between biology and engineering, with the goal of producing renewable fuels from sunlight at efficiencies and costs that complement, rather than compete with, established PV and wind technologies. For the future of integrated renewable energy systems, both natural and artificial photosynthesis will play important roles alongside conventional solar electricity generation.

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DISCLOSURE OF AI USAGE

This paper was produced with the assistance of generative artificial intelligence tools. Perplexity AI (Deep Research) was used for literature search and source identification. Claude (Anthropic, claude-sonnet-4-6) was used to support paper structuring, drafting, and L^AT_EX typesetting. All sources were identified and provided by the author; factual content was verified against the cited references. The final paper reflects the intellectual work and judgment of the author.

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